Chapter 12, Problem 3P

Problem

Given a balanced Y-connected three-phase generator with a line-to-line voltage of $\mathbf{V}_{ab} = 100\underline{/45^\circ}\ \mathrm{V}$ and $\mathbf{V}_{bc} = 100\underline{/165^\circ}\ \mathrm{V}$, determine the phase sequence and the value of V*ca*.

Step-by-step solution

Step 1 of 2

Consider the given values.

$V_{ab} = 100\angle 45^\circ\,\mathrm{V}$ and $V_{bc} = 100\angle 165^\circ\,\mathrm{V}$.

The voltages are in negative or *acb* sequence since the voltage V_{bc} leads V_{ab} by 120°.

Compare $V_{ab} = 100\angle 45^\circ\,\mathrm{V}$ with $V_L\angle\theta^\circ$.

The values of V_L and θ° are $100\,\mathrm{V}$ and 45° respectively.

Step 2 of 2

Calculate the value of the voltage V_{ca}.

$$\begin{aligned} V_{ca} &= V_L\angle\theta^\circ - 120^\circ \\ &= 100\angle 45^\circ - 120^\circ \\ &= 100\angle -75^\circ\,\mathrm{V} \end{aligned}$$

Substitute 45° for θ° and $100\,\mathrm{V}$ for V_L.

$$\begin{aligned} V_{ca} &= 100\angle 45^\circ - 120^\circ \\ &= 100\angle -75^\circ\,\mathrm{V} \end{aligned}$$

Therefore, the phase voltage V_{ca} is, $\boxed{100\angle -75^\circ\,\mathrm{V}}$.